

How to Strategize Human Content Creation in the Era of GenAI?

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Abstract

Generative AI (GenAI) will have significant impact on content creation platforms. In this paper, we study the dynamic competition between a GenAI and a human contributor. Unlike the human, the GenAI's content only improves when more contents are created by the human over time; however, GenAI has the advantage of generating content at a lower cost. We study the algorithmic problem in this dynamic competition model about how the human contributor can maximize her utility when competing against the GenAI for content generation over a set of topics. In time-sensitive content domains (e.g., news or pop music creation) where contents' value diminishes over time, we show that there is no polynomial time algorithm for finding the human's optimal (dynamic) strategy, unless the randomized exponential time hypothesis is false. Fortunately, we are able to design a polynomial time algorithm that naturally cycles between myopically optimizing over a short time window and pausing and provably guarantees an approximation ratio of $\frac{1}{2}$. We then turn to time-insensitive content domains where contents do not lose their value (e.g., contents on history facts). Interestingly, we show that this setting permits a polynomial time algorithm that maximizes the human's utility in the long run. Finally, we conduct simulations that demonstrate the advantage of our algorithms in comparison to a collection of baselines.

CCS Concepts

• Theory of computation → Algorithmic game theory.

Keywords

Generative AI, Hardness and Approximation Algorithms

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1 Introduction

The emergence of Generative Artificial Intelligence (GenAI) has ushered in a transformative era across numerous practical domains, including text [26], image [29], and video generation [2, 8, 16]. As AI-generated content proliferates on online platforms, ranging from media-sharing (e.g., Instagram [11] and TikTok [1]) to news outlets, the implications for human content creators are profound.

The rise of GenAI presents both challenges and opportunities for human content creators. While GenAI poses a competitive threat due to its capacity for rapid and cost-effective content generation, it also relies on human-created data for training and refinement. This interdependence creates a strategic interplay where human creators may adapt their content strategies to differentiate themselves from AI-generated outputs — a theme that this work aims to address.

A key concern surrounding GenAI is its potential to displace human creators due to its lower production costs and its capability to generate useful content as well as human creators. However, the imperative for *timely* and accurate content in online platforms necessitates a continuous influx of human-generated material to serve as training data for GenAI models, for example, media contents on hot topics and recent news comments. This reliance on human input suggests a cyclical relationship where the absence of human content could degrade the quality of AI-generated outputs, leading to user dissatisfaction and a renewed demand for human creativity. In this scenario, human creators may experience a resurgence in value as their content garners increased attention and engagement from users, which creates an intrinsic difficulty to analyze the *dynamics* between human and GenAI competition. To capture this *time-sensitive* nature of content generation, we discount the value of human contents when we model the capability of GenAI as a function of human created contents. Naturally, the older the contents, the more its value is discounted.

Notably, there are also situations where the value of contents is *time-insensitive*, e.g., historical facts, scientific knowledge and professional skills training. For these domains, through our model we uncover the threat that GenAI will replace human content creation can be real. We model such time-insensitive scenarios by assuming that the quality of human-generated contents is not discounted over time. For each topic, there is a moment that human content creation will eventually stop. To optimize utility in this model, one important consideration of the human creator is to delay their time to be driven out by GenAI and meanwhile maximize their accumulated rewards before GenAI's content creation becomes as competitive.

1.1 Our Contributions

Our main technical contributions are fourfold: (1) In Section 2, we introduce the first model to study dynamic competition between a human agent and a GenAI agent for content creation across different topics in an online platform. Our model introduces key elements that capture advantages and disadvantages of both human and GenAI agent. In addition, the model enables us to associate with each topic a measure of “information value” at current time by discounting the contents according to how recently they were created. Different topics have different discounting factors: e.g., time-sensitive contents are more discounted than insensitive ones. Our goal is to design algorithms/strategies that strategize the human agent’s choice of content topics in order to maximize the human’s utility.

(2) In Section 3, we study the setting where topics are time-sensitive and therefore the (human) contents used for GenAI training for all topics are discounted. We start by proving that the problem is computationally hard. Specifically, we show that finding an optimal strategy for the human cannot be done in polynomial time unless the randomized exponential time hypothesis is false. Perhaps surprisingly, this intractability remains true even in simple special settings, hence illustrating the fundamental barriers for the human to strategize in such highly dynamic environments. We then introduce a novel polynomial time algorithm that alternates between running a short-term optimal strategy and pausing (not generating any content) and show that it guarantees an approximation ratio close to $\frac{1}{2}$. We further prove that it is necessary for any algorithm in time-sensitive setting to pause for a portion of the time horizon as otherwise the algorithm will provably obtain a vacuous approximation of the optimal utility in some situations.

(3) In Section 4, we consider the setting where topics are time-insensitive and therefore the contents are not discounted (i.e., their information value to GenAI will not reduce over time). We pay special interest to long-term interactions due to a desire of understanding the human’s long term utility. For time-insensitive topics (e.g., facts or scientific knowledge), a corollary from our model is that the human player will eventually find it not profitable to create any new contents due to GenAI’s increasing capability, hence the human will “exit” the platform at some round. However, how to maximize total utility for the human before exiting the platform is a highly non-trivial algorithmic question. We start with a simple baseline algorithm that runs in exponential time. Through careful analysis about the problem structure, we are able to improve the algorithm, leading to a polynomial time algorithm that maximizes the human utility. This setting provides an interesting contrast to the discounted setting where optimizing utility is computationally hard.

(4) In Section 5, we carry out experiments that demonstrate the superiority of our algorithms from Sections 3 and 4 over a collection of baselines.

A few remarks regarding this work’s contributions are worth mentioning. As one of the first studies of such dynamic human-vs-GenAI content creation competition, our model is simple and basic. It captures the essential aspects of the dynamic competition but admittedly does not capture its whole landscape. On one hand, as we show even this simple setup is already highly non-trivial

for algorithm development. On the other hand, we do see various possibilities to extend our basic model formulation which cannot be all covered in a single work. Thus, we believe our algorithmic results offer useful insights for future studies of richer and concrete problem settings. Finally, we present intuitions for all technical results in the main body but defer some proofs to the appendix due to space limits.

1.2 Additional Related Work

There has been a recent surge of interest in studying online content economy and competition among different content creators [4, 5, 17, 18, 20, 35–37, 39]. More related to us are the works that aim to understand content creators’ behaviors, for example, how creators will specialize at the equilibrium [20], how creators’ strategic behavior affects social welfare [35], and how to design optimization method for long-term welfare considering content creators’ strategic behaviors [4, 5, 18, 19, 25, 36, 39]. Our work differ from these studies in a few key aspects. Most importantly, the previous works study the competition among only human creators whereas our model is set to study a more macro level question and features the competition between humanity collectively represented as a single Human player and technology represented as a single GenAI player.

Recently, a few papers have studied the interaction between GenAI and human content creators. In particular, [32] study the social impact from GenAI on content creators in question answering websites such as Stack Overflow. Their main focus is on designing algorithms that maximize revenue for the GenAI whereas we consider the optimization problem from the perspective of the human and accordingly our objectives and focus is very different. Moreover, [27] study content diversity/homogeneity when content creators use GenAI as a tool showing that GenAI usage could lead to less diverse content. Additionally, Yao et al. [37] study the competition between human and GenAI. However, their model is at a more micro-level with many human creators who either compete with a single GenAI or use GenAI as an additional content creation tool.

2 A Model of Dynamic Human-vs-GenAI Content Competition

Basic Setup. Motivated by content creation competition to attract Internet user traffic on recommender systems (e.g., Instagram, TikTok, Youtube, etc.), in this section we introduce a basic model to capture the dynamic competition between human content creators and GenAI content generation. There are k different topics (e.g., sports, politics, or nature), denoted by $[k] = \{1, \dots, k\}$, for which humans and GenAI can create contents. For convenience, we shall refer to each topic as an *arm*. Additionally, and naturally, there is always an “opt out” option for the human to exit the game and receive 0 utility. In this paper we focus on the dynamic two-agent competition between a single human agent and a single GenAI agent. This model can be interpreted as the competition between all humans collaboratively as one team and all GenAIs collectively as another team. The goal of our work is to study the human team’s optimal competition strategy against GenAIs. While a more complex model could consider multiple self-interested humans and

GenAIs, we start our study from the two-agent setting because: (1) even such two-agent competition is already highly non-trivial hence worth a thorough investigation; (2) human creators' distinct content creation behavior used to be primarily caused by their different creation costs [36], but this factor becomes negligible when facing GenAIs with order-of-magnitude lower creation cost. Thus this assimilates human creators' behaviors due to facing the same strong GenAI competitor, effectively making humans as if in a team. The human agent's content creation capability for each arm/topic $i \in [k]$ is described by two values c_i, μ_i where c_i is normalized to be within $[0, 1]$ is the human's cost of creating a topic- i content and μ_i captures user satisfaction or *reward* for a human-created topic- i content.¹

Modeling the Capability of GenAIs. We consider repeated content creation over rounds $1, 2, \dots, T$. Let arm $b_t \in [k]$ denote the topic choice by the human agent at round t , whereas the GenAI chooses an arm $a_t \in [k]$. As mentioned above, the reward resulting from the human's pull of arm b_t is μ_{b_t} . However, the reward $\tilde{\mu}_{a_t}$ resulting from the GenAI's pull requires more careful modeling and will naturally depend on the capability of the GenAI technology. Motivated by extensive recent studies of the GenAI's capability as a function of the number of training data (also widely known as *scaling laws* [14, 21]), we assume $\tilde{\mu}_{a_t}$ is a function of the total (possibly discounted) number of times that arm a_t has been pulled by the human. Notably, we assume GenAI's pulling of an arm does not improve the GenAI's capability on this arm; this is consistent with the state-of-the-art GenAI technologies which are all trained on high-quality human-created data but not synthetic GenAI contents. Formally, let

$$N_i(t) = \sum_{s=1}^t (\gamma_i)^{t+1-s} \times \mathbb{I}(b_s = i) \quad (1)$$

denote the total discounted number of times that the human pulled arm $i \in [k]$ at round t . Here the discounting factor $\gamma_i \in [0, 1]$ is a parameter that captures the time-sensitivity of a topic i and can generally be different for different topics. For instance, news topics are time-sensitive and will have small γ_i because earlier data are less useful for generating today's news content² whereas factual knowledge (e.g., history events, linear algebra, etc.) is less time-sensitive and will have large γ_i . We will pay special interest to the cases where all topics have a discounted number of pulls $\gamma_i < 1, \forall i \in [k]$ in Section 3 and where the value of contents is not discounted $\gamma_i = 1, \forall i \in [k]$ in Section 4.

Hence for any arm $i \in [k]$, we model the reward $\tilde{\mu}_i$ of the GenAI's pull of arm i as $\tilde{\mu}_i(t) := \mu_i - g_i(N_i(t-1))$ where $g_i(\cdot)$ is the quality *gap* between human-created contents and GenAI-created ones for arm i and is a function of the total number of human-created contents. We also refer to g_i as the *shrinkage* function. Besides naturally assuming g_i is positive and decreasing (i.e., more human-created contents, less gap), we do not assume any concrete form for g_i . We allow each arm $i \in [k]$ to have its own

¹Internet users' reward for a content may have noise in general but this will be modeled later when describing the users' (random) choice between human-created and GenAI-created contents.

²Currently, one potential recipe for such time-sensitive topics with low γ_i could be using retrieval-augmented generation (RAG) [24, 31] that feeds the news articles as prompts/contexts to generate. This improvement could still be captured by our model using an increased γ_i value.

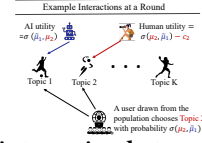


Figure 1: Example interaction between the human, GenAI, and Internet user at a given round.

shrinkage function $g_i(\cdot)$, capturing the fact that the GenAI's content generation capabilities may differ across topics. Note that $\tilde{\mu}_i$ depends (only) on human's historical arm pulls. For convenience, we refer to $\tilde{\mu}_i(t) = \mu_i - g_i(N_i(t-1))$ as the *generative mean* of arm i at round t . We assume the cost for GenAI to create a topic- i content is smaller than human's cost. Without loss of generality, we *normalize* the GenAI's content creation cost to be 0 (i.e., human's cost $c_i \in [0, 1]$ should be interpreted as the relative cost). Finally, we make a minor technical assumption that the rewards of all (human or GenAI) created contents are bounded in $[\mu_l, \mu_u]$. I.e., $\forall i \in [k] : \mu_u \geq \mu_i \geq \mu_i - g_i(T) \geq \mu_l \geq 0$.

Competitions and Utilities. We are now ready to model the dynamic Human-vs-GenAI competition. Figure 1 shows a visualization of the interaction. In each round t an Internet user arrives and faces two contents, one created by human with user reward r_{b_t} and another created by GenAI with reward $\tilde{r}_{a_t}$. Following the standard choice model, we assume the user chooses one of the arms b_t, a_t randomly with probabilities specified by a *link function* σ . Importantly, this means that the user is choosing between only the current round's contents. This is realistic since frequent recommender system users often only consume fresh contents as older contents have already been viewed (this is also why these systems often recommend fresh contents anyway). Let $i \succ j$ denote the event that the user selects arm i over j . The probability of this event is given by the link function $\sigma(\cdot, \cdot)$, i.e., $\mathbb{P}(i \succ j) = \sigma(\mu_i, \mu_j) \in [0, 1]$. Naturally, $\mathbb{P}(j \succ i) = 1 - \sigma(\mu_i, \mu_j) = \sigma(\mu_j, \mu_i)$. Besides the standard (and natural) assumptions that $\sigma(\cdot, \cdot)$ is monotonically increasing in the first variable and decreasing in the second, our results do not rely on any concrete form of the link function σ . We also allow the human to possibly not select an arm in any round t , and denote this action by $\emptyset$. In that case, the human incurs no cost but GenAI's content will be chosen with probability 1. That is, $\sigma(\emptyset, \mu) = 0 = 1 - \sigma(\mu, \emptyset)$ for any $\mu \in [\mu_l, \mu_u]$.

The objective of both the human and GenAI is to maximize the user traffic of their contents, defined as the expected number of user pulls. Recall that the GenAI's generation costs have been normalized to 0. Hence the competition leads to the following accumulated utilities for each agent.

$$\text{Human Agent's Accumulated Utility: } u_T(\mathbf{b}; \mathbf{a}) = \sum_{t=1}^T [\sigma(\mu_{b_t}, \tilde{\mu}_{a_t}(t)) - c_{b_t}] \quad (2)$$

$$\text{GenAI Agent's Accumulated Utility: } u_T^{\text{AI}}(\mathbf{a}; \mathbf{b}) = \sum_{t=1}^T \sigma(\tilde{\mu}_{a_t}(t), \mu_{b_t}) \quad (3)$$

where $\mathbf{b} = (b_1, \dots, b_T)$ is the realized sequence of the human's actions, similarly $\mathbf{a} = (a_1, \dots, a_T)$ is the realized sequence of the

GenAI's actions. At times, we may need to refer to the agents' accumulated utilities between rounds t_1 and t_2 ($t_1 \leq t_2$, both inclusive). In this case, we use $u_{t_1:t_2}$ and $u_{t_1:t_2}^{\text{AI}}$ to denote these utilities which are defined similarly as in Equation (2) and (3) but with summation from t_1 to t_2 . Note that if the human or GenAI randomize their actions then the utilities are the expectations of (2) and (3) for the human and GenAI, respectively.

The accumulated utility functions in Equation (2) and (3) induce a dynamic competition between the human and the GenAI. The two agents' interactions hinge on GenAI's generative mean $\tilde{\mu}_{a_t}(t) = \mu_{a_t} - g_{a_t}(N_{a_t}(t-1))$ of arm a_t , which depends on the human's past actions.

We make a few remarks about the link function σ . User's probabilistic choice is motivated by reward noise, which is standard in choice models [6, 12]. That is, both $\mu_{b_t}, \tilde{\mu}_{a_t}(t)$ should be viewed as expected rewards of the contents whereas the realized true reward should be $\mu_{b_t} + \epsilon_1, \tilde{\mu}_{a_t}(t) + \epsilon_2$. The randomness in user choice comes from the random noise term ϵ_1, ϵ_2 . For instance, if they are logit distributions, we arrive at the standard Bradley-Terry model with $\sigma(\mu_{b_t}, \tilde{\mu}_{a_t}(t)) = \frac{e^{\mu_{b_t}}}{e^{\mu_{b_t}} + e^{\tilde{\mu}_{a_t}(t)}}$. If they are i.i.d Gaussian noise with zero mean and σ^2 variance, then $\sigma(\mu_{b_t}, \tilde{\mu}_{a_t}(t))$ equals the probability that Gaussian $\mathcal{N}(\mu_{b_t} - \tilde{\mu}_{a_t}(t), 2\sigma^2)$ is positive [38].

Final Problem Formulation and Preliminary Model Analysis. Standing in the human's shoes, we study the human's optimal dynamic competition with GenAI. One last key modeling piece is the content creation strategy of the GenAI. In the present work, we consider the competition with GenAI that has myopic content creation strategies as formalized in the following assumption.

Assumption 2.1. *At a given round t the GenAI selects the arm with the current highest generative mean $\hat{a}_t = \max_{i \in [k]} \tilde{\mu}_i(t)$ to maximize the probability of being chosen by the user for this round.*

It is straightforward to prove that selecting $\hat{a}_t$ maximizes the probability of being chosen at round t . If we denote by $p_t(i)$ and $q_t(j)$ the probability that the human and the GenAI will select arms i and j in round t , respectively, then we have $\mathbb{P}(a_t > b_t) = \sum_{i,j \in [k]} p_t(i) q_t(j) \sigma(\tilde{\mu}_i, \mu_j) = \sum_{i \in [k]} q_t(i) \left(\sum_{j \in [k]} p_t(j) \sigma(\tilde{\mu}_i, \mu_j) \right) \leq \sum_{j \in [k]} p_t(j) \sigma(\tilde{\mu}_{\hat{a}_t}, \mu_j)$, where the last step follows from the mean monotonicity of σ .

Let vector $\mathbf{a}^* \in [k]^T$ denote the sequence of GenAI's arm choices. Notably, $\mathbf{a}^*$ is not fixed and is a function of the human pulls $\mathbf{b}$. To emphasize this dependence, we also write $\mathbf{a}^*(\mathbf{b})$. Hence, this gives rise to our final dynamic optimization problem for the human:

Human's Dynamic Optimization Problem:

$$\max_{\mathbf{b} \in [k]^T} u_T(\mathbf{b}; \mathbf{a}^*(\mathbf{b})) \quad (4)$$

where $\hat{a}_t = \max_{i \in [k]} \tilde{\mu}_i(t)$

In Appendix B, we also present an equivalent formulation of the above dynamic optimization problem as searching for the longest path on a certain "state transition graph". This formulation will be useful in constructing some technical arguments. We conclude this section with some preliminary analysis about the above optimization problem. We start by illustrating the behavioral assumption of the GenAI, as detailed in Assumption 2.1. Recall that our goal

is to optimize human's utility in the competition with GenAI. The following result shows that optimizing against the GenAI strategy in Assumption 2.1 is equivalent to robust optimization of human's worst-case utility.

Proposition 2.2. *For any $\mathbf{b}$ and $\mathbf{a}$, we have $u_T(\mathbf{b}; \mathbf{a}^*(\mathbf{b})) \leq u_T(\mathbf{b}; \mathbf{a})$.*

PROOF. Note that strategy $\mathbf{b}$ could possibly be randomized (i.e., randomizing over the selected arms), by invoking the monotonicity of σ we have $u_T(\mathbf{b}; \mathbf{a}^*(\mathbf{b})) = \mathbb{E}[\sum_{t=1}^T [\sigma(\mu_{b_t}, \tilde{\mu}_{a_t^*}) - c_{b_t}]] \leq \mathbb{E}[\sum_{t=1}^T [\sigma(\mu_{b_t}, \tilde{\mu}_{a_t}) - c_{b_t}]] = u_T(\mathbf{b}; \mathbf{a})$ $\square$

Proposition 2.2 implies that by assuming GenAI strategy $\mathbf{a}^*$, the human can establish a guarantee on her worst-case utility even if GenAI does not follow the strategy at the end. This offers a justification of GenAI's strategy assumption through the lens of robust optimization.

Next we offer an alternative view of the GenAI's strategy through a game-theoretic lens. Notably, if the GenAI is forward-looking, its strategy generally could have been much more complex since it may intentionally pull sub-optimal arms at times in order to encourage human content creation on certain arms so that it would benefit from this creation in the future. Such a strategic consideration will lead to a general dynamic game between the GenAI and the human, and finding an equilibrium in such dynamic games is notoriously hard [15, 22, 23]. However, our following result shows that, if the human's strategy is independent of the GenAI's choices, the GenAI strategy in Assumption 2.1 is actually an optimal GenAI strategy over the entire horizon. Formally, we say $\mathbf{b}$ is *oblivious*, if each b_t is a function only of its own history $b_1, \dots, b_{t-1}$.³

Proposition 2.3. *If the human uses an oblivious strategy $\mathbf{b}$ then the GenAI myopic strategy $\mathbf{a}^*(\mathbf{b})$ as described in Assumption 2.1 is also an optimal forward-looking strategy that maximizes its total utility over the entire time horizon.*

PROOF. We denote the human's oblivious strategy by $\mathbf{b}^{\text{obv}}$. Let $\mathbf{a}'$ denote some arbitrary GenAI strategy. Note that both $\mathbf{b}^{\text{obv}}$ and $\mathbf{a}'$ could possibly be randomized. We follow a proof by induction and accordingly start with the base case (utility over round 1) which follows immediately by the monotonicity of σ .

Base Case: $u_{1:1}^{\text{AI}}(\mathbf{a}^*; \mathbf{b}^{\text{obv}}) \geq u_{1:1}^{\text{AI}}(\mathbf{a}'; \mathbf{b}^{\text{obv}})$.

Inductive Step: Note that for any two strategies (even if randomized) $\mathbf{b}$ for the human and $\mathbf{a}$ for the GenAI, by the linearity of the expectations the utility brakes down as

$$u_{1:t}^{\text{AI}}(\mathbf{a}; \mathbf{b}) = u_{1:t-1}^{\text{AI}}(\mathbf{a}; \mathbf{b}) + \mathbb{P}(a_t > b_t; \mathbf{a}; \mathbf{b}) \quad (5)$$

Since the human's strategy is oblivious then it is independent of the GenAI's strategy and therefore the probability that the human will pull arm i at round t will be $p_t(i, \mathbf{b}^{\text{obv}})$ for both strategies $\mathbf{a}^*$ and $\mathbf{a}'$. Therefore, it follows by Assumption 2.1 that $\mathbb{P}(a_t^* > b_t; \mathbf{a}^*, \mathbf{b}^{\text{obv}}) \geq \mathbb{P}(a_t > b_t; \mathbf{a}', \mathbf{b}^{\text{obv}})$. Further, since $u_{1:t-1}^{\text{AI}}(\mathbf{a}^*; \mathbf{b}^{\text{obv}}) \geq u_{1:t-1}^{\text{AI}}(\mathbf{a}'; \mathbf{b}^{\text{obv}})$ by the inductive hypothesis then $u_{1:t}^{\text{AI}}(\mathbf{a}^*; \mathbf{b}^{\text{obv}}) \geq u_{1:t}^{\text{AI}}(\mathbf{a}'; \mathbf{b}^{\text{obv}})$. By finally setting $t = T$ the proposition is complete. $\square$

³This can be equivalently viewed as the human committing to a long-horizon strategy $\mathbf{b}$ at the beginning of the competition, after which the GenAI best responds.

Finally, a minor concern one might have is whether a randomized human strategy $\mathbf{b}$ could be strictly beneficial for the human since randomization has been proven to be powerful in strategic plays. Our next lemma shows that the answer is No.

Lemma 2.4. *Randomization is not more beneficial: there always exists an optimal strategy for the human that is deterministic.*

The proof of the Lemma is an application of the method of conditional expectations (see, e.g. [33]) so we omit it. Essentially, by considering an optimal randomized strategy at round $t = T$ we can always derandomize it for $t = T$ by selecting the arm that gives the maximum utility for $t = T$. By recursively invoking this argument for $t = T - 1, T - 2, \dots, 1$ we arrive at a deterministic strategy.

For the rest of the paper we will focus on finding the optimal solution $\mathbf{b}^*$ to Problem (4). Since $\mathbf{a}^*(\mathbf{b})$ is determined by $\mathbf{b}$, we will drop the argument $\mathbf{a}^*$ for the GenAI strategy when it is clear in the context and simply write $u_T(\mathbf{b})$ for $u_T(\mathbf{b}; \mathbf{a}^*)$.

3 Time-Sensitive Domains: $\gamma < 1$

We start our algorithmic study for time-sensitive content domains, in which the relevance or usefulness of the content diminishes over time (e.g., news or fashion contents). We capture this domain by considering discounting factors γ_i 's that are all bounded below 1, i.e., $\forall i \in [k], \gamma_i \leq \gamma_{\max} < 1$ for some $\gamma_{\max}$. Based on Lemma 2.4 we restrict our analysis to deterministic strategies. In Subsection 3.1 we prove the computational hardness of our problem. Specifically, unless the well-believed Randomized Exponential Time Hypothesis (rETH) is false [9]⁴, we cannot maximize the human's utility in polynomial time. In fact, this intractability holds even in simpler special situations when all arms have the same cost and, additionally, have the same discounting factor or the same shrinkage function. Therefore, in Subsection 3.2 we design a polynomial time approximation algorithm.

3.1 The Hardness of Human Utility Maximization

The hardness proof for our problem is based on a connection to the blocking bandits problem of Basu et al. [3]. Like in the standard stochastic bandit setting, in blocking bandits we have k arms each having its own mean μ_i . However, each arm also has a delay value D_i which is a positive integer. If an arm i is pulled at round t then it cannot be pulled again for $D_i - 1$ many consecutive rounds after t , making $t + D_i$ the first subsequent round where i maybe pulled again. Interestingly, Basu et al. [3] show that even when the delay values D_i and means μ_i of each arm are known, the pure maximization problem⁵ of selecting the arms to maximize the total accumulated rewards through the horizon subject to satisfying the delay constraints is computationally hard. Specifically, they show that the problem does not admit a pseudo-polynomial time algorithm in the number of arms unless the randomized exponential time hypothesis is false. We prove the hardness of our problem by reducing a special instance of blocking bandits (shown to be computationally hard) to the problem of finding a deterministic

optimal strategy in our discounted contents setting. Our main result is as follows:

Theorem 3.1. *Let $\gamma_i < 1, \forall i \in [k]$, unless the randomized exponential time hypothesis (rETH) is false, there is no polynomial time algorithm for finding a deterministic optimal strategy for the human even if all arms have the same cost c and the same discounting factor γ or if all arms have the same cost c and the same shrinkage function $g(\cdot)$.*

The key idea in the reduction is to make *violating the delay* D_i for an arm i costly by making the resulting generative mean too high and therefore leading to small utility. Based on the delay value D_i for an arm i we essentially will either tune its shrinkage function $g_i(\cdot)$ or its discount factor γ_i so that the final generative mean is too high once the delay is violated but remains below a certain threshold if the delay is not violated. Therefore, the only way to maximize utility is to pull the arms but without violating their delay. The full details of the proof however require tuning a large number of parameters as well as careful bounding arguments leading to a tedious derivation.

3.2 An (Almost) 1/2-Approximation

To establish our bounds, we make natural and standard assumptions about our setting. Specifically, we assume that all shrinkage functions have a Lipschitz constant upper bound L_g , i.e., $\forall i \in [k] : |g_i(n_1) - g_i(n_2)| \leq L_g |n_1 - n_2|$. Similarly, we assume that the link function is also L_σ Lipschitz, i.e., $|\sigma(\mu, \mu_1) - \sigma(\mu, \mu_2)| \leq L_\sigma |\mu_1 - \mu_2|$. Further, we assume that the instances we consider have a lower bound on the maximum utility we can gain in the first round, i.e., $\max_{i \in [k]} (\sigma(\mu_i, \tilde{\mu}_{a_i^*}) - c_i) \geq u_0 > 0$.⁶

Our approximation algorithm follows a natural structure of cycling between a short window (interval) where it pulls arms based on a *myopically* optimal strategy and an equally sized window where it pauses (does not pull arms). Before introducing our algorithm, we start with a useful observation. Denote by **Myopic-OPT**(τ) the optimal value we obtain by running a strategy which optimizes utility from the first round $t = 1$ to round $t = \tau$. The following lemma states that the utility we obtain over this window is larger than the utility we obtain from a full horizon optimal strategy (optimizing until $t = T$) over any equally sized window.

Lemma 3.2. *Let $\mathbf{b}^* \in [k]^T$ be a deterministic optimal strategy for the human, then for any τ consecutive rounds from any t to $t + (\tau - 1)$ we have **Myopic-OPT**(τ) $\geq u_{t:t+\tau-1}(\mathbf{b}^*)$.*

PROOF. Suppose we follow the same strategy as $\mathbf{b}^*$ from rounds t and ending at $t + (\tau - 1)$ but in the first τ rounds, i.e. starting from round 1 to round τ to optimize utility for the first τ rounds, we will call this strategy $\mathbf{b}'$. Then it follows that for any arm $i \in [k]$ the new discounted number of pulls $N'_i(r)$ satisfies $N'_i(r) \leq N_i(r + t - 1)$, $\forall r \in \{1, 2, \dots, \tau\}$. Denote the maximum generative mean by $\tilde{\mu}_{a_r^*}$ and $\tilde{\mu}_{a_r'}$ for strategies $\mathbf{b}'$ and $\mathbf{b}^*$, respectively. Further, we denote by b'_r and b_r the arms pulled at round r by $\mathbf{b}'$ and $\mathbf{b}^*$, respectively.

⁴rETH is a popular assumption in complexity theory and widely believed to be true [7, 10, 13, 28, 34].

⁵By pure here we mean that no learning is involved and optimization is done with all parameters known.

⁶Note this is a mild assumption as it can be established by assuming the instances have a lower bound on the maximum mean, the shrinkage function at value 0, and an upper bound on the cost. In fact, the computationally hard instances in the proof of Theorem 3.1 fall under this category.

Then we have the following bound: $\sum_{r=1}^{\tau} \sigma(\mu_{b_r}, \tilde{\mu}'_{a_r}) - c_{b_r} \geq \sum_{r=1}^{\tau} \sigma(\mu_{b_r}, \tilde{\mu}_{a_r}^*) - c_{b_r} = \sum_{r=t}^{t+\tau-1} \sigma(\mu_{b_r}, \tilde{\mu}_{a_r}^*) - c_{b_r} = u_{t:t+\tau-1}(\mathbf{b}^*)$.

Where the above follows since for all $r \in \{1, 2, \dots, \tau\}$ we must have $\tilde{\mu}'_{a_r} \leq \tilde{\mu}_{a_r}^*$ since $N'_i(r) \leq N_i(r+t-1)$. Therefore, we have $\mathbf{Myopic-OPT}(\tau) \geq \sum_{r=1}^{\tau} \sigma(\mu_{b_r}, \tilde{\mu}'_{a_r}) - c_{b_r} \geq u_{t:t+\tau-1}(\mathbf{b}^*)$. $\square$

It is easy to see that an algorithm with running time exponential in τ (specifically, $O(k^\tau)$) can be easily designed (e.g., by simply running a single source longest path algorithm in a directed acyclic graph as detailed in Appendix B) for finding the optimal human strategy $\mathbf{b}$ for $\mathbf{Myopic-OPT}(\tau)$. We denote this strategy by $\mathbf{b}_{\mathbf{Myopic}}(\tau)$. Although the run-time is exponential in τ , when τ is a small constant (as will be in our design), this leads to a polynomial time algorithm for finding $\mathbf{b}_{\mathbf{Myopic}}(\tau)$.

Using $\mathbf{b}_{\mathbf{Myopic}}(\tau)$, we construct the strategy MYOPICALLY-OPTIMIZE-THEN-PAUSE as shown in Algorithm 1 which alternates between running the same actions from $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ and pausing for τ rounds. Theorem 3.3 shows that we can obtain a $\frac{1-\epsilon}{2}$ approximation if we set $\tau = \left\lceil \log_{\left(\frac{1}{\gamma_{\max}}\right)} \left(\frac{\gamma_{\max}}{(1-\gamma_{\max})^2} \cdot \frac{L_g L_\sigma}{u_o} \cdot \frac{1}{\epsilon} \right) \right\rceil = O(\log \frac{1}{\epsilon})$. Note that this does not require τ to have a dependence on the input parameters k or T . The key intuition behind our algorithm is that pausing would enable us to essentially force the GenAI to “forget” the contents since their value will be discounted. Therefore, when we run $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ again we would obtain a large utility due to Lemma 3.2. However, since we are pausing for only a finite duration, the number of contents does not truly go back to 0. Therefore, the core technical challenges in the proof lie at analyzing the utility loss during the pause window and the utility degradation during the $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ window.

Algorithm 1 MYOPICALLY-OPTIMIZE-THEN-PAUSE

- 1: Set $\tau = \left\lceil \log_{\left(\frac{1}{\gamma_{\max}}\right)} \left(\frac{\gamma_{\max}}{(1-\gamma_{\max})^2} \cdot \frac{L_g L_\sigma}{u_o} \cdot \frac{1}{\epsilon} \right) \right\rceil$
 - 2: Find the optimal human strategy for the first τ rounds and denote it by $\mathbf{b}_{\mathbf{Myopic}}(\tau)$.
 - 3: **while** $t \leq T$ **do**
 - 4: Alternate between following strategy $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ and pausing for τ many rounds.
 - 5: **end while**
-

Theorem 3.3. Suppose $\gamma_{\max}, L_g, L_\sigma, u_o$ are constants, then MYOPICALLY-OPTIMIZE-THEN-PAUSE (Algorithm 1) runs in $O(T + k^{O(\log \frac{1}{\epsilon})})$ and achieves an approximation ratio of $\frac{1-\epsilon}{2} \cdot \frac{\lfloor \frac{T}{2\tau} \rfloor}{\lfloor \frac{T}{2\tau} \rfloor + 1}$.

PROOF. Before we give the proof of the theorem, we establish the following lemma. Define n_ϵ to be

$$n_\epsilon = \frac{\gamma_{\max}}{1 - \gamma_{\max}} \cdot \gamma_{\max}^\tau \quad (6)$$

Since we set $\tau = \left\lceil \log_{\left(\frac{1}{\gamma_{\max}}\right)} \left(\frac{\gamma_{\max}}{(1-\gamma_{\max})^2} \cdot \frac{L_g L_\sigma}{u_o} \cdot \frac{1}{\epsilon} \right) \right\rceil$, the following lemma can be established.

Lemma 3.4.

$$\frac{L_g L_\sigma}{1 - \gamma_{\max}} n_\epsilon \leq \epsilon \mathbf{Myopic-OPT}(\tau) \quad (7)$$

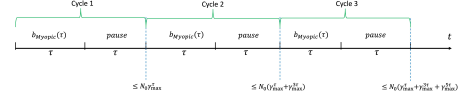


Figure 2: Illustration of the upper bound on the discounted number of contents at the end/beginning of each cycle.

PROOF. $\frac{L_g L_\sigma}{1 - \gamma_{\max}} n_\epsilon = L_g L_\sigma \cdot \frac{\gamma_{\max}}{(1 - \gamma_{\max})^2} \cdot \gamma_{\max}^\tau \leq L_g L_\sigma \cdot \frac{\gamma_{\max}}{(1 - \gamma_{\max})^2} \cdot \frac{(1 - \gamma_{\max})^2}{\gamma_{\max}} \cdot \frac{u_o}{L_g L_\sigma} \cdot \epsilon = u_o \epsilon \leq \epsilon \mathbf{Myopic-OPT}(\tau)$. Where we used the Fact that $\mathbf{Myopic-OPT}(\tau) \geq \mathbf{Myopic-OPT}(1) \geq u_o$ in the last step. $\square$

Note that the algorithm cycles between running $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ and pausing for τ many rounds. Let i_0 be the arm that has the maximum number of discounted pulls at the end of round τ when running $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ and denote that number of pulls by N_0 . We will show that at the beginning of any cycle the number of pulls for any arm will be small. Specifically, we have

Claim 3.5. At the beginning of any cycle at round t we have $\forall i \in [k]$ we have $N_i(t-1) \leq n_\epsilon$.

PROOF. Arms only receive pulls when $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ is active. It follows that after the end of $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ in the first cycle we have

$$N_0 \leq \sum_{r=1}^{\tau} \gamma_{\max}^r = \frac{\gamma_{\max}}{1 - \gamma_{\max}} \cdot (1 - \gamma_{\max}^\tau) \leq \frac{\gamma_{\max}}{1 - \gamma_{\max}} \cdot (1 - \gamma_{\max}^{2\tau}) \quad (8)$$

At the beginning of the m^{th} cycle we would have an upper bound for the discounted number of pulls of (see Figure 2 for an illustration) would be: $N_0 \cdot (\gamma_{\max}^\tau + \gamma_{\max}^{3\tau} + \gamma_{\max}^{5\tau} + \dots) = N_0 \sum_{\ell=0}^{\infty} \gamma_{\max}^{\tau(2\ell+1)} = N_0 \frac{\gamma_{\max}^\tau}{1 - \gamma_{\max}^{2\tau}} \leq \frac{\gamma_{\max}}{1 - \gamma_{\max}} \cdot \gamma_{\max}^\tau \leq n_\epsilon$. Where last step follows from the definition of n_ϵ in (6). $\square$

Now we establish that $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ is approximately optimal if $\forall i \in [k]$ we have $N_i(t) \leq n_\epsilon$.

Claim 3.6. Suppose we run the $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ starting from a state where the number of pulls for all arms is at most n_ϵ then the utility we obtain is at least $(1 - \epsilon) \cdot \mathbf{Myopic-OPT}(\tau)$.

PROOF. We will add a prime ' to denote a quantity when the number of pulls is starting from at most n_ϵ , if it starts from 0 then we omit the prime '. If we follow strategy $\mathbf{b}_{\mathbf{Myopic}}(\tau)$ then at any round r in the start of the cycle we would have $N'_i(r) \leq N_i(r) + n_\epsilon \gamma^{r-1}$. It follows by the Lipschitz property of $g_i(\cdot)$ and the fact that it is a decreasing function that $g_i(N'_i(r)) \geq g_i(N_i(r)) - L_g n_\epsilon \gamma^{r-1}$. Therefore, the maximum generative mean increases by at most $L_g n_\epsilon \gamma^{r-1}$, i.e. $\tilde{\mu}'_{a_r} \leq \tilde{\mu}_{a_r} + L_g n_\epsilon \gamma^{r-1}$. Further, it follows by the Lipschitz property of σ that in the first half of the cycle the utility is: $\sum_{r=1}^{\tau} \sigma(\mu_{b_r}, \tilde{\mu}'_{a_r}) - c_{b_r} \geq \sum_{r=1}^{\tau} \sigma(\mu_{b_r}, \tilde{\mu}_{a_r} + L_g n_\epsilon \gamma^{r-1}) - c_{b_r} \geq \left[\sum_{r=1}^{\tau} \sigma(\mu_{b_r}, \tilde{\mu}_{a_r}^*) - c_{b_r} \right] - \left[\sum_{r=1}^{\tau} L_g L_\sigma n_\epsilon \gamma^{r-1} \right] \geq \left[\sum_{r=1}^{\tau} \sigma(\mu_{b_r}, \tilde{\mu}_{a_r}^*) - c_{b_r} \right] - L_g L_\sigma n_\epsilon \frac{1}{1 - \gamma_{\max}} \geq \mathbf{Myopic-OPT}(\tau) - \epsilon \cdot \mathbf{Myopic-OPT}(\tau) = (1 - \epsilon) \mathbf{Myopic-OPT}(\tau)$. Where in the step before the last used Lemma 3.4. $\square$

The strategy runs $\mathbf{b}_{\text{Myopic}}(\tau)$ for a window of τ where it obtains a $(1 - \epsilon)$ approximation for that window then it pauses for another window of size τ . Therefore, its approximation ratio over the entire horizon T is at least $\frac{1-\epsilon}{2} \frac{\lfloor \frac{T}{2\tau} \rfloor}{\lfloor \frac{T}{2\tau} \rfloor + 1}$.

The run-time to find $\mathbf{b}_{\text{Myopic}}(\tau)$ using the exponential DAG of Appendix B is $O(k^\tau) = O(k^{O(\log \frac{1}{\epsilon})})$ and it takes $O(\tau) = O(1)$ memory to save the arm pulls of $\mathbf{b}_{\text{Myopic}}(\tau)$. Further, since the algorithm has to make T many decisions for the arm pulls over the whole horizon, we obtain a run-time of $O(T + k^{O(\log \frac{1}{\epsilon})})$. $\square$

Clearly, as $T \rightarrow \infty$ the approximation ratio is $\frac{1-\epsilon}{2}$. Further, since τ is increasing in $\gamma_{\max}$, smaller values for $\gamma_{\max}$ lead to smaller τ and a better approximation ratio for a fixed T .

Our algorithm pauses for a large portion of the horizon. Although this seems counter-intuitive, pausing is actually necessary for securing high utilities. Specifically, by constructing a special example we obtain the below theorem which shows that not pausing can lead to negative utility.

Theorem 3.7. *There exists an instance such that the human will receive negative utility if she pulls arms for more than $0.1T$ rounds (not necessarily consecutively).*

PROOF. Let us assume that the human pulls arms for αT rounds where $\alpha \geq 0.1$. Consider the following instance where we have two arms, i.e., $k = 2$ and $\mu_1 = \mu_2 = \mu$, $g_1(\cdot) = g_2(\cdot) = g(\cdot)$, $c_1 = c_2 = c$, and $\gamma_1 = \gamma_2 = \gamma = \frac{1}{2} - \epsilon$ where $\epsilon \in (0, \frac{1}{2})$.

Since $\gamma = \frac{1}{2} - \epsilon$, then the following is always true for all $\kappa \geq 2$:

$$\gamma^{\kappa+1} \frac{1}{1-\gamma} < \gamma^\kappa \quad (9)$$

Claim 3.8. *The discounted number of pulls if an arm was played in the last κ rounds is at least γ^κ and if the arm has not been played in the last κ rounds then the number of pulls is at most $\gamma^{\kappa+1} \frac{1}{1-\gamma}$.*

PROOF. The first part is immediate. The second part holds since if the arm is not played in the last κ rounds then the maximum number of pulls at the round preceding the current round by $\kappa + 1$ is at most $\frac{1}{1-\gamma}$ and therefore if we do not play the arm for $\kappa + 1$ rounds then the discounted number of pulls is at most $\gamma^{\kappa+1} \frac{1}{1-\gamma}$. $\square$

The set of rounds where arms are played can be decomposed into $\mathcal{T}_{\text{less}}$ and $\mathcal{T}_{\text{more}}$ which are the set of rounds where an arm is pulled after waiting for less than κ and κ rounds or more, respectively. Clearly, we have:

$$\alpha \cdot T = |\mathcal{T}_{\text{less}}| + |\mathcal{T}_{\text{more}}| \quad (10)$$

Note that $|\mathcal{T}_{\text{more}}| \leq \frac{1}{\kappa+1}T$ since once if an arm is pulled then it must be that we have at least $\kappa + 1$ rounds where no arm was played. Therefore, we can lower bound $|\mathcal{T}_{\text{less}}|$ as follows $|\mathcal{T}_{\text{less}}| = \alpha \cdot T - |\mathcal{T}_{\text{more}}| \geq \alpha \cdot T - \frac{1}{\kappa+1}T = (\alpha - \frac{1}{\kappa+1})T$.

Now we set $\kappa = \argmin_{r \in \mathbb{Z}_{>0}} \frac{1}{r+1} < \alpha$. Let $N_l = \gamma^{\kappa+1} \frac{1}{1-\gamma}$ and let $N_u = \gamma^\kappa$. Note that $N_u > N_l$ by (9). We set the utility values to the following:

$$\sigma(\mu, \tilde{\mu}_{a_t^*}) - c = v > 0 \text{ if } N_l(t-1), N_2(t-1) \leq N_l \quad (11)$$

$$\sigma(\mu, \tilde{\mu}_{a_t^*}) - c = v' \text{ if } N_l(t-1) \text{ or } N_2(t-1) \geq N_u \quad (12)$$

We set the values of v and v' such that

$$v' = \frac{-\beta}{(\alpha(\kappa+1)-1)}v \quad (13)$$

where $\beta > 1$. From the above the utility is at most $v' \cdot |\mathcal{T}_{\text{less}}| + v \cdot |\mathcal{T}_{\text{more}}| \leq \frac{-\beta}{(\alpha(\kappa+1)-1)}v(\alpha - \frac{1}{\kappa+1})T + v \frac{1}{\kappa+1}T = vT \frac{1}{\kappa+1}(1-\beta) < 0$. Since $\beta > 1$ then the above utility is negative.

Now we prove the second part. To do that we only need to establish a lower bound on the utility of the optimal algorithm OPT . Consider an algorithm that pulls arm 1 and then pauses for κ rounds. Then it must achieve a utility of at least $v \lfloor \frac{T}{\kappa+1} \rfloor$. Therefore, $\text{OPT} \geq v \lfloor \frac{T}{\kappa+1} \rfloor > 0$. It follows since the utility obtained by the algorithm is negative that the multiplicative approximation ratio is less than zero. $\square$

4 Time-Insensitive Domains: $\gamma = 1$

We now turn to time-insensitive domains, where contents remain relevant, accurate, and useful regardless of when they are accessed. This captures content generation for timeless literature, history, science, and math knowledge, etc. We capture this situation by assuming all arms have discounting factors equal to 1: $\forall i \in [k] : \gamma_i = 1$. Clearly, the more pulls an arm receives by the human the higher the value of its generative mean $\tilde{\mu}_i(t)$. Since there is no discounting in this setting, the generative means $\tilde{\mu}_i$'s will be non-decreasing and in the long run GenAI will be equally competitive as the human for every topic, i.e., $\tilde{\mu}_i \rightarrow \mu_i, \forall i \in [k]$. For our algorithmic result, the concrete format of the g_i functions and link function σ do not matter. What matters are the following “thresholding” values for the generative mean and number of pulls, which are determined by g_i , σ , and c_i .

Definition 4.1 (Exit Mean and Exit Pull). For an arm $i \in [k]$, we define $\mu_{i,\text{exit}}$ as the smallest value such that $\sigma(\mu_i, \mu_{i,\text{exit}}) - c_i < 0$. Further, the smallest number n such that $\mu_i - g_i(n) \geq \mu_{i,\text{exit}}$ is called the *exit pull* and denoted by $n_{i,\text{exit}}$. We assume that for each i we have $n_{i,\text{exit}} < \infty$.

In other words, the $n_{i,\text{exit}}$ 'th pull of arm i is the first time i has a generative mean larger than $\mu_{i,\text{exit}}$ and hence the first time a negative utility will be generated if both the human and GenAI pull i since $\sigma(\mu_i, \mu_{i,\text{exit}}) - c_i < 0$. The assumption $n_{i,\text{exit}} < \infty$ implies any arm has such a moment.

Throughout this section, we naturally assume that the human will never pull an arm if it has negative utility, since it is surely worse than the trivial 0-utility opt-out action even when the long-term utility is considered. We first characterize the set of “plausible” arms that can possibly be pulled, as summarized in the following observation.

Observation 4.2. Let $\mathcal{P}_a^*(t)$ denote the set of arms at round t that have non-negative human utilities. Then we have $\mathcal{P}_a^*(t) = \{i \in [k] \mid \mu_{i,\text{exit}} > \tilde{\mu}_{a_t^*}\}$. Moreover, $\mathcal{P}_a^*(t) \subseteq \mathcal{P}_a^*(t-1), \forall t \geq 2$.

To develop some intuition, it is useful to understand why the observation holds. First, any $i \in \mathcal{P}_a^*(t)$ must give non-negative utility. Indeed, when pulling i , the human's utility $\sigma(\mu_i, \tilde{\mu}_{a_t^*}) - c_i$ is at least 0 because $\tilde{\mu}_{a_t^*} < \mu_{i,\text{exit}}$ and $\mu = \mu_{i,\text{exit}}$ is the smallest value to make $\sigma(\mu_i, \mu) - c_i < 0$. Second, any $i \notin \mathcal{P}_a^*(t)$ must have negative

human utility because such i satisfies $\tilde{\mu}_{a_t^*} \geq \mu_{i,\text{exit}}$, hence pulling i leads to human utility $\sigma(\mu_i, \tilde{\mu}_{a_t^*}) - c_i \leq \sigma(\mu_i, \mu_{i,\text{exit}}) - c_i < 0$. Finally, $\mathcal{P}_{a^*}(t) \subseteq \mathcal{P}_{a^*}(t-1)$ simply because $\tilde{\mu}_{a_t^*}$ is non-decreasing in t whereas $\mu_{i,\text{exit}}$ does not change for each i . As a consequence, we know that $\mathcal{P}_{a^*}(1)$ contains all the arms that can possibly have positive utilities for human. Now we define the long-horizon setting that we focus on in order to understand humanity's long-term welfare.

Definition 4.3 (Long Horizon Setting). The horizon T is long if $T \geq \sum_{i \in \mathcal{P}_{a^*}(1)} n_{i,\text{exit}}$.

Notably, when the horizon is long, the human will eventually have negative utilities at every arm. Since each arm i can be pulled at most $n_{i,\text{exit}}$ times as the human never pulls an arm with negative utility due to the existence of 0-utility opt-out choice, and only arms in $\mathcal{P}_{a^*}(1)$ can possibly be pulled by the human. Consequently, under the long horizon setting the human's optimization reduces to collecting as much utility as possible before being fully driven out from the competition.⁷ Despite appearing frustrating at first glance, this should be expected to hold in the long run for time-insensitive content domains such as history knowledge or basic math where creating good content is costly for the human whereas GenAI is becoming increasingly more efficient. However, maximizing the human's utility in long horizon settings is highly non-trivial. In contrast to the computational hardness of Section 3, we maximize utility in polynomial time for this setting as shown in the Theorem below.

Theorem 4.4. *In the long horizon setting, there is an $O(Tk^3)$ algorithm for maximizing the human's utility.*

We note that it is not difficult to show that simple baseline algorithms would either run in exponential time or not lead to an optimal strategy. Through a collection of lemmas we define a class of strategies which we call *synchronizing* strategies that essentially have the special property of pulling an arm whenever pulling it would not increase the maximum generative mean value $\tilde{\mu}_{a_t^*}$ at round t . The simpler structure of synchronizing strategies is the core reason behind our polynomial $O(Tk^3)$ run-time.

5 Experiments

We compare the performance of our algorithms to a collection of baselines through simulations. The link function σ is set to the standard Bradley-Terry model [6], i.e., $\sigma(\mu_i, \mu_j) := \frac{e^{\mu_i}}{e^{\mu_i} + e^{\mu_j}}$. Further, the shrinkage function for a given topic i is set to $g_i(N) := q_i / \sqrt{N + (\frac{q_i}{h_i})^2}$ where q_i and h_i essentially control the vertical scaling and horizontal shift of the function with $g_i(0) = h_i$. It follows that the Lipschitz constants would be $L_\sigma = \frac{1}{4}$ and $L_g = \max_{i \in [k]} \frac{h_i^3}{2q_i^2}$.

5.1 Time-Sensitive Domains

Here we test the performance of our **Myopic-then-Pause** (Algorithm 1) against a collection of baselines. We consider the **Greedy** baseline where at each round t the human pulls the utility-maximizing arm, i.e., $b_t = \arg\max_{i \in [k]} (\sigma(\mu_i, \tilde{\mu}_{a_t^*}) - c_i)$. We also consider a "pure" myopic algorithm (**Pure-Myopic**) where the **Myopic-then-Pause**

⁷Note that this is different from staying within the competition for as long as possible since longest stay may not necessarily lead to maximum accumulated utility.

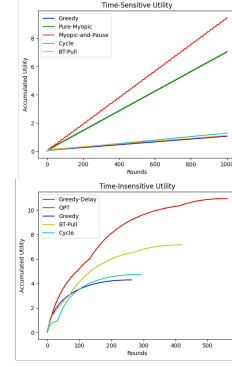


Figure 3: Accumulated utility vs round for each algorithm in: (top) time-sensitive domain. (bottom) time-insensitive domain.

algorithm is run without an enforced pausing interval. Furthermore, we consider a fourth algorithm **BT-Pull** based on the Bradley-Terry choice model where the human randomly pulls the arms according to their means — specifically, at round t an arm i is pulled with probability $p_i = \frac{j_t(i)\mu_i}{\sum_{i' \in [k]} j_t(i')\mu_{i'}}$, where $j_t(i) = 1$ if pulling arm i at round t would result in nonnegative utility and 0 otherwise. If all arms result in negative utility, the human instead pauses for one round. We consider a final baseline **Cycle** where the human cycles between arms, skipping arms that would result in negative utility, and pausing if all arms would have resulted in negative utility. For this experiment, we have 4 arms ($k = 4$) and means $\mu = [.73, .85, .9, .95]$, discounting factors $\gamma = [.5, .48, .47, .45]$, costs $c = [.53, .56, .59, .58]$, shrinkage parameters $q = [.1, .2, .5, .2]$, and $h = [.3, .45, .6, .6]$. The horizon is set to $T = 1,000$, and the approximation constant for **Myopic-then-Pause** is set to $\epsilon = 0.1$. Figure 3 (top) shows the total accumulated utility at a given round t , i.e., $u_{1,t}$ for our algorithm and baselines. Clearly, we outperform the other baselines by a significant margin.

5.2 Time-Insensitive Domains

Our algorithm which we call **OPT** for this setting is from Theorem 4.4. We include some of the baselines from the time-sensitive domain setting. We still compare against **Greedy**, **BT-Pull**, and **Cycle**. In addition, another baseline is **Greedy-Delay** where in round t the human pulls the arm that gives the smallest increase in the GenAI's maximum generative mean. At an intuitive level, this baseline would enable the human to stay longer before exiting the platform where gaining non-negative utility is impossible. We test an instance with 4 arms setting the means to $\mu = [.86, .95, .87, .95]$, costs to $c = [.56, .52, .55, .53]$, and shrinkage parameters $q = [1.4, 1.3, 1.5, 1.8]$ and $h = [.75, .8, .6, .78]$. To ensure that we are in the long horizon setting we set $T = 561$. From Figure 3 (bottom) clearly, our algorithm achieves the best performance significantly outperforming **Greedy**, **BT-Pull**, and **Cycle**.

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A Useful Mathematical Fact

We have the following useful fact. Since the proof follows standard arguments, it is omitted.

Lemma A.1. For $D \geq 2$ and $\gamma \in [0, \frac{1}{2}]$ we have $\gamma \cdot \frac{1}{1-\gamma^D} < \gamma + \gamma^D$.

B Utility Maximization as Longest Path

We show how the human’s dynamic optimization problem of (4) is equivalent to the longest path problem in a state transition graph. Before doing that we introduce the following notation for the generative mean of an arm. Specifically, we let $\tilde{\mu}_i(p = r) = \mu_i - g_i(r)$, i.e., the generative mean value of arm i when the human has pulled the arm r many times (or in general r is the discounted number of pulls when $\gamma_i < 1$). Note that we include “ $p =$ ” in the argument of $\tilde{\mu}_i(p = r)$ to emphasize that it is a function of the number of pulls. As stated earlier the action of not pulling an arm will be represented by pulling arm $\emptyset$. From standard graph theory, it is known that finding the longest path in a directed acyclic graph (DAG) from a single source can be done in $O(|V| + |E|)$ run-time where V is the set of vertices and E is the set of edges (see, e.g. [30]).

This leads us to construct the state transition graph as follows:

- (1) **Node Attributes:** Each node will have $k + 1$ many numbers $(n_1, n_2, \dots, n_k; d)$ associated with it. For all $i \in [k]$, n_i denotes the

number of (discounted) pulls for arm i . Attribute d indicates its distance from the root (first node). The first node has $n_1 = n_2 = \dots = n_k = 0$ and $d = 0$. (2) **Edge Connections:** Given a node $(n_1, n_2, \dots, n_k; d)$ such that $d < T$ for each $i \in [k]$ draw an edge to a new node $(n'_1, n'_2, \dots, n'_k; d')$ such that $n'_i = \gamma_i + \gamma_i^2 \cdot n_i$ and $\forall j \in [k], j \neq i : n'_j = \gamma_j^2 \cdot n_j$. Further, for the new node set $d' = d + 1$. Moreover, set the edge weight to $\sigma(\mu_i, \max_{j \in [k]} \tilde{\mu}_j(p = n_j)) - c_i$. Finally, draw an edge to a new no action $\emptyset$ node, with $n'_i = \gamma_i^2 n_i, \forall i \in [k]$ and set the edge weight to 0 and $d' = d + 1$. If $d = T$, then the node has no outgoing edges. See Figure 4 for an example of this state transition graph. The graph is clearly a directed acyclic graph (DAG). It follows directly that maximizing utility is equivalent to finding the longest path starting from the first node ($n_1 = 0, n_2 = 0, \dots, n_k = 0; d = 0$). The graph has a node for each state that can be reached and therefore a size of $\Omega(k^T)$ and would take $O(k^T)$ time to construct, however, it will be important in our discussion, and we will refer to it as the *exponential graph*.

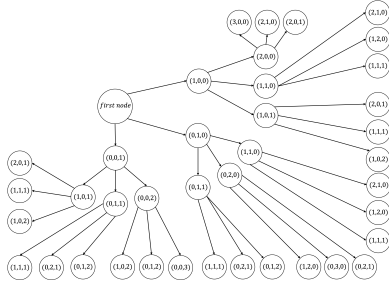


Figure 4: An example of the exponential graph for $k = T = 3$ and $\gamma_i = 1, \forall i \in [k]$. Attribute d and no action $\emptyset$ nodes are not drawn for simplicity.

C Omitted Proofs

Proof of Theorem 3.1 from Subsection 3.1

In blocking bandits [3], each arm $i \in [k]$ has a mean μ_i and additionally has associated with it a *delay value* $D_i \geq 1$. If arm i is pulled at round t then it cannot be pulled again until at least round $t + D_i$. In the MAXREWARD problem, the mean and delay values are known and the objective is to pull arms through the horizon to maximize the accumulated expected rewards while not violating the delay values. From Basu et al. [3] we have the following:

Lemma C.1 ([3]). *Suppose that we have $k + 1$ arms where the first k arms have a mean of 1 and a delay value D_i with $\sum_{i=1}^k \frac{1}{D_i} = 1$, the last arm has a mean 0 and a delay of 1, then deciding if MAXREWARD has a solution of value of at least T cannot be done in pseudo-polynomial time in the number of arms k unless the randomized exponential time hypothesis is false.*

The reduction will be from this instance. Since the lemma above shows that no pseudo-polynomial time algorithm in the number of arms k exists, we will further assume that for all arms $i \in [k]$ we have $D_i = \text{poly}(k)$ where $\text{poly}(\cdot)$ is some polynomial function. We now give Lemma C.2 that proves the first part of Theorem 3.1.

Lemma C.2. *Suppose $\gamma_i < 1$ for all $i \in [k]$, unless the randomized exponential time hypothesis is false there does not exist a polynomial time deterministic optimal strategy for the human even if all arms have the same cost c and the same discount factor γ .*

PROOF. We will have $k + 1$ arms. The means of the first k arms is set to the same value of $\mu = 0.9$ whereas the last will have a value of $\mu_{k+1} = 0.1$. Further, all arms have the a cost of $c = 0.6$ and a discount factor $\gamma = \frac{1}{2}$. Now, we have the following Lemma which establishes a bound between violating and non-violating pulls.

Lemma C.3. *For any arm $i \in [k]$ with $D_i \geq 2$ if at round $t - 1$ the delay rule was violated then $N_i(t - 1) \geq \gamma + \gamma^{D_i} > \frac{\gamma}{1 - \gamma^{D_i}}$. On the other hand, if the delay rule was not violated at any round up to and including $t - 1$ then we have $N_i(t - 1) \leq \frac{\gamma}{1 - \gamma^{D_i}} < \gamma + \gamma^{D_i}$.*

PROOF. First, by an immediate application of Lemma A.1 we have for $\gamma = \frac{1}{2}$ that $\gamma \cdot \frac{1}{1 - \gamma^{D_i}} < \gamma + \gamma^{D_i}$.

Now, suppose the pull violates the delay rule at round t then it must be the case that arm i was pulled at least D_i rounds before t . This means that the smallest value for the discounted number of contents will be $N_i(t) \geq \gamma + \gamma^{D_i}$. See Figure 5 for an example.

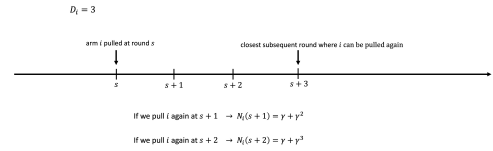


Figure 5: Example illustrating the value for the discounted number of contents for an arm i with $D_i = 3$ if its delay value is violated.

On the other hand, if we are at round t and the pulls for arm i did not violate the delay then the highest value for the discounted number of contents would be as follows (see Figure 6 for an illustration): $N_i(t) \leq \gamma + \gamma^{D_i+1} + \gamma^{2D_i+1} + \gamma^{3D_i+1} + \dots = \sum_{\ell=0}^{\infty} \gamma^{D_i\ell+1} = \gamma \cdot \frac{1}{1 - \gamma^{D_i}}$ □

Now we complete the details of the reduction, the generative mean for an arm i is $\tilde{\mu}_i(t) = \mu_i - g_i(N_i(t - 1))$. Note that since $\gamma = \frac{1}{2}$, the highest discounted number of contents will be $\frac{\gamma}{1 - \gamma} = 1$. For the last arm $g_{k+1}(n) = 0.1 - 0.1n$. On the other hand, for the first k arms we set $g_i(n) = g(\lambda_i n), \forall i \in [k]$, where $\lambda_i > 0$ is to be set according to the delay value D_i . The function $g(\cdot)$ (same for all $i \in [k]$) is set equal to the following: $g(n) = 0.7 - 0.35n$. Let $p^* = \frac{1}{2}$. Now for any $i \in [k]$ we will set λ_i as follows: $\lambda_i = \frac{p^*}{\gamma + \gamma^{D_i}}$.

This implies that if the delay rule for an arm i is violated then in the next round we have $\lambda_i N_i(t) \geq p^*$, see Figure 7 for an illustration.

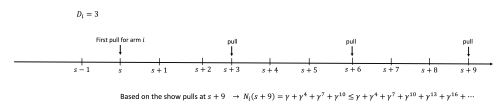


Figure 6: Example illustrating the upper bound on the discounted number of contents for an arm i with $D_i = 3$ if its delay value is not violated.

Critically we note that since $D_i = \text{poly}(k)$, then λ_i can be computed in polynomial time and saved in polynomial space. Based on the above, the following fact holds.

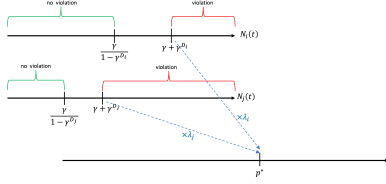


Figure 7: The figure shows how when the delay value is violated for any arm i its coefficient λ_i would lead to $\lambda_i N_i(t) \geq p^*$.

Fact C.4. For any arm $i \in [k]$ with $D_i \geq 2$ if at round $t-1$ the delay rule was violated then $\tilde{\mu}_i(t) \geq \tilde{\mu}_u = 0.375$. On the other hand, if the delay rule was not violated at any round up to and including $t-1$ then we have $\tilde{\mu}_i(t) \leq \tilde{\mu}_l < \tilde{\mu}_u$ where $\tilde{\mu}_l = 0.9 - g(\max_{i \in [k]} \frac{p^*}{\gamma + \gamma^{D_i}} \frac{\gamma}{1 - \gamma^{D_i}})$.

PROOF. If the delay is violated at round $t-1$ for an arm $i \in [k]$ then we have in the next round t by Lemma C.3 that $\tilde{\mu}_i(t) = \mu_i - g_i(N_i(t-1)) \geq \mu_i - g(\lambda_i \cdot (\gamma + \gamma^{D_i})) = 0.375 = \tilde{\mu}_u$. On the other hand if the delay rule is not violated then the maximum possible value given to the function $g(\cdot)$ can be $\max_{i \in [k]} \lambda_i \frac{\gamma}{1 - \gamma^{D_i}} = \max_{i \in [k]} \frac{p^*}{\gamma + \gamma^{D_i}} \frac{\gamma}{1 - \gamma^{D_i}} < p^*$. Therefore, based on the definition of $\tilde{\mu}_l$ in the fact, we will always have $\tilde{\mu}_i(t) \leq \tilde{\mu}_l < \tilde{\mu}_u$. $\square$

Based on the mean and shrinkage functions, the maximum generative mean value is at least 0.2 and at most 0.55. Further, the human will choose arms of mean either 0.9 (first k) or 0.1 (the last arm). Therefore, it is sufficient to define the link function over $[0.1, 0.9] \times [0.1, 0.9]$. We will now set the value for link function as follows: (A) if $\mu_1 = 0.1$ then $\sigma(0.1, \mu_2) = \frac{1 + (0.1 - \mu_2)}{2}$. (B) if $\mu_1 = 0.9$ then we set the link function as

$$\sigma(0.9, \mu_2) = \begin{cases} \frac{1 + (0.9 - \mu_2)}{2} & \text{if } \mu_2 \geq \tilde{\mu}_l \\ \frac{1 + (0.9 - \tilde{\mu}_l)}{2} & \text{if } \mu_2 < \tilde{\mu}_l \end{cases} \quad (14)$$

(C) for $\mu_1 \in (0.1, 0.9)$ the link function is as follows

$$\sigma(\mu_1, \mu_2) = \begin{cases} \frac{1}{2}(\mu_1 - 0.1) + \sigma(0.1, \mu_2) & \text{if } \mu_1 \leq \mu_2 \\ \frac{\sigma(0.9, \mu_2) - \frac{1}{2}}{0.9 - \mu_2}(\mu_1 - \mu_2) + \frac{1}{2} & \text{if } \mu_1 > \mu_2 \end{cases} \quad (15)$$

Clearly, σ is continuous, $\sigma(\mu, \mu) = \frac{1}{2}$ and that it is increasing in the first argument. Note that by our choice for the shrinkage functions, the GenAI will always choose an arm from the first k set since it is straightforward to show that $\tilde{\mu}_{a_t^*} \geq \tilde{\mu}_i(t) \geq 0.2 > 0.1 \geq \tilde{\mu}_{k+1}(t)$, $\forall i \in [k]$ and $\forall t \in [T+1]$. Therefore, we can upper bound the utility of the last arm of mean μ_{k+1} by $\sigma(\mu_{k+1}, \tilde{\mu}_{a_t^*}) - c \leq \frac{1 + (0.1 - 0.2)}{2} - 0.6 = -0.15 < 0$. Therefore, the last arm always gives negative utility.

From the above we can now separate the utility we gain from the first k arms based on whether we violated their delay rule or not

Fact C.5. If the delay rule for any arm $i \in [k]$ with $D_i \geq 2$ is violated at round $t-1$ then the utility at the next round will be at most $\theta_l - c$ where $\theta_l = \frac{1 + (0.9 - \tilde{\mu}_u)}{2}$. On the other hand, if the delay rule is not violated then we can obtain a maximum utility of exactly $\theta_u - c > 0$ where $\theta_u = \frac{1 + (0.9 - \tilde{\mu}_l)}{2} > \theta_l$.

Now that we have given the details of the reduction. We can immediately establish that the human can obtain a utility value of $(\theta_u - c) \cdot (T+1)$ if and only if he always pulls an arm from the set of first k arms without violating their delay values except possibly at the last round $T+1$. This implies a reduction from MAXREWARD and completes the proof of Lemma C.2. $\square$

The proof of Theorem 3.1 can be completed by a Lemma similar to Lemma C.2 with the shrinkage function being the same instead of the discount factor. It follows by a similar proof so we omit it.

Proof of Theorem 4.4 from Section 4

Lemmas C.6 and C.8 show that a deterministic optimal strategy with simple special properties exists enabling us to reduce the size of the graph to a polynomial one.

Lemma C.6. There exists a deterministic optimal strategy which always pulls arms from $i \in \mathcal{P}_a^*(t)$ when $|\mathcal{P}_a^*(t)| \geq 1$ and stops playing if $|\mathcal{P}_a^*(t)| = 0$.

PROOF. The second part of the statement is proved immediately since when $|\mathcal{P}_a^*(t)| = 0$ only negative utility can be accumulated. The first part can be proved using the following claim.

Claim C.7. Consider a deterministic strategy which at round t pulls an arm $i \notin \mathcal{P}_a^*(t)$ when $|\mathcal{P}_a^*(t)| \geq 1$ then it can be replaced by a strategy which pulls the same arms before round t , pulls an arm from $\mathcal{P}_a^*(t)$ in round t , and achieves at least the same utility.

PROOF. Let the followed strategy be denoted by $\mathbf{b}$. If $\mathbf{b}$ does not pull an arm from $\mathcal{P}_a^*(t)$ at round t when $|\mathcal{P}_a^*(t)| \geq 1$ then either (1) no arm from $\mathcal{P}_a^*(t)$ is pulled again until the end of the horizon or (2) there exists some arm in $\mathcal{P}_a^*(t)$ that was not pulled at round t and that will be pulled after round t . For case (1), if we deviate to a strategy $\mathbf{b}'$ which pulls the same arms as $\mathbf{b}$ up to round $t-1$, then pulls some arm $j \in \mathcal{P}_a^*(t)$ at round t and then stops playing we would obtain a higher utility since: $u_T(\mathbf{b}') = u_{1:t-1}(\mathbf{b}) + (\sigma(\mu_j, \tilde{\mu}_{a_t^*}) - c_j) > u_{1:t-1}(\mathbf{b}) + u_{t:T}(\mathbf{b}) = u_T(\mathbf{b})$. Note that $u_{t:T}(\mathbf{b}) < 0$. For case (2), let arm $j \in \mathcal{P}_a^*(t)$ be the first arm in $\mathcal{P}_a^*(t)$ to be pulled after round t and let that round when it is pulled be $t_j \leq T$. We will deviate to a strategy which pulls the same arms as $\mathbf{b}$ up to round $t-1$, then pulls j at round t and then pulls the same arms pulled by $\mathbf{b}$ from round $t_j + 1$ to T and then stops playing. It follows by the above that for a round $r \in \{t+1, \dots, T - (t_j - t)\}$ strategy $\mathbf{b}'$ pulls the arm as pulled by $\mathbf{b}$ in round $r + (t_j - t)$, see Figure 8 for an illustration. Further, for any round r denote by $\tilde{\mu}_{a_r^*}$ and $\tilde{\mu}'_{a_r^*}$ the maximum generative mean at that round r under strategies $\mathbf{b}$ and $\mathbf{b}'$, respectively. Note that by construction of $\mathbf{b}'$ we have $\tilde{\mu}'_{a_r^*} \leq \tilde{\mu}_{a_r^*}, \forall r \in [T]$. Now we can lower bound the utility under $\mathbf{b}'$ as follows: $u_T(\mathbf{b}') = u_{1:t-1}(\mathbf{b}) + (\sigma(\mu_j, \tilde{\mu}'_{a_t^*}) - c_j) + \sum_{r=t+1}^{T-(t_j-t)} [\sigma(\mu_{b_r}, \tilde{\mu}'_{a_r^*}) - c_{b_r}] \geq u_{1:t-1}(\mathbf{b}) + (\sigma(\mu_j, \tilde{\mu}_{a_t^*}) - c_j) +$

$$\sum_{r=t_j+1}^T [\sigma(\mu_{b_r}, \tilde{\mu}_{a_r^*}) - c_{b_r}] = u_{1:t-1}(\mathbf{b}) + u_{t_j:T}(\mathbf{b}) > u_{1:t-1}(\mathbf{b}) + u_{t:t_j-1}(\mathbf{b}) + u_{t_j:T}(\mathbf{b}) = u_T(\mathbf{b})$$



Figure 8: Illustration of the arm pulls under strategy $\mathbf{b}$ and the deviation strategy $\mathbf{b}'$ for case (2) of Lemma C.6.

By (recursively) invoking the above claim on any strategy which does not pull an arm $i \notin \mathcal{P}_{a^*}(t)$ when $|\mathcal{P}_{a^*}(t)| \geq 1$ we can obtain a new strategy which achieves at least the same utility that always pulls arms from $\mathcal{P}_{a^*}(t)$. $\square$

At a given round t define $\mathcal{H}_{a^*}(t) = \{i \in \mathcal{P}_{a^*}(t) \mid \tilde{\mu}_i(p = N_i(t-1) + 1) \leq \tilde{\mu}_{a_t^*}\}$, i.e. the set of arms which if pulled would not increase the value of the maximum generative mean. The following lemma states that if the maximum generative mean increases because of a pull, then an optimal deterministic strategy would then “synchronize” the generative means of the other arms to make them equal to the current maximum generative mean or slightly below it.

Lemma C.8. *In the long horizon setting there exists an optimal deterministic strategy which at a given round t would always pull an arm $i \in \mathcal{H}_{a^*}(t)$ when $|\mathcal{H}_{a^*}(t)| \geq 1$.*

PROOF. Since we are in the long horizon setting ($T \geq \sum_{i \in \mathcal{P}_{a^*}(1)} n_{i,\text{exit}}$) then by Lemma C.6 there exists a deterministic optimal strategy which always pulls arms from $\mathcal{P}_{a^*}(t)$ and exits (stops playing) at some round $\tau \leq T$. Now consider the following claim:

Claim C.9. *Consider a deterministic strategy which at round t pulls an arm $i \notin \mathcal{H}_{a^*}(t)$ when $|\mathcal{H}_{a^*}(t)| \geq 1$, then it can be replaced by a strategy which pulls the same arms before round t , pulls an arm from $\mathcal{H}_{a^*}(t)$ in round t , and achieves at least the same utility.*

PROOF. Suppose that there exists $j \in \mathcal{H}_{a^*}(t)$ that was not pulled under strategy $\mathbf{b}$ at round t then either (1) j is pulled in some following round $t_j > t$ or (2) it is not pulled again. Consider case (1) where j is pulled again. We will deviate to strategy $\mathbf{b}'$ which pulls the same arms as $\mathbf{b}$ up to round $t-1$, then pulls j at round t and then in the rounds from $t+1$ to t_j pulls the same arms as pulled by $\mathbf{b}$ from t to t_j-1 in the same order, then finally pulls the same arms as $\mathbf{b}$ from rounds t_j+1 to τ as shown in Figure 9. The utility using $\mathbf{b}'$ can be lower bounded as follows: $u_T(\mathbf{b}') = u_{t-1}(\mathbf{b}) + (\sigma(\mu_j, \tilde{\mu}_{a_t^*}) - c_j) + \sum_{r=t+1}^{\tau} [\sigma(\mu_{b_r}, \tilde{\mu}_{a_r^*}) - c_{b_r}] \geq u_{t-1}(\mathbf{b}) + \sum_{r=t}^{t_j} [\sigma(\mu_{b_r}, \tilde{\mu}_{a_r^*}) - c_{b_r}] + \sum_{r=t_j+1}^{\tau} [\sigma(\mu_{b_r}, \tilde{\mu}_{a_r^*}) - c_{b_r}] = u_T(\mathbf{b})$.

Now we consider case (2) where arm j is never pulled again. Since we are in the long horizon setting, i.e. $T \geq \sum_{i \in \mathcal{P}_{a^*}(1)} n_{i,\text{exit}}$ it follows that $\tau < T$. To see that, suppose $\tau = T$ then since we assume that $\mathbf{b}$ is an optimal strategy then $\tau \leq \sum_{i \in \mathcal{P}_{a^*}(1)} n_{i,\text{exit}}$ and since $T \geq \sum_{i \in \mathcal{P}_{a^*}(1)} n_{i,\text{exit}}$ then $\tau = \sum_{i \in \mathcal{P}_{a^*}(1)} n_{i,\text{exit}}$. The only way for an optimal strategy to have $\tau = \sum_{i \in \mathcal{P}_{a^*}(1)} n_{i,\text{exit}}$ is to pull each arm $i \in \mathcal{P}_{a^*}(1)$, $n_{i,\text{exit}}$ many times. Since $j \in \mathcal{H}_{a^*}(t)$ is not pulled at or after round t it follows that j was not pulled $n_{j,\text{exit}}$ many times and therefore we must have $\tau < \sum_{i \in \mathcal{P}_{a^*}(1)} n_{i,\text{exit}} \leq T$.

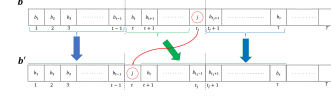


Figure 9: Illustration of the arm pulls under strategy $\mathbf{b}$ and the deviation strategy $\mathbf{b}'$ for case (1) of Lemma C.8.

Now that we established that $\tau < T$, the deviation strategy $\mathbf{b}'$ is simple. Specifically, $\mathbf{b}'$ pulls the same arms as $\mathbf{b}$ up to round $t-1$, then it pulls j at round t and then from rounds $t+1$ to $\tau+1$ it pulls the same arms $\mathbf{b}$ pulls from rounds t to τ in the same order. Since $\sigma(\mu_j, \tilde{\mu}_{a_t^*}) - c_j \geq 0$ and pulling j does not increase the maximum generative mean it follows that $u_T(\mathbf{b}') \geq u_T(\mathbf{b})$. $\square$

By the invoking the above claim recursively we can obtain a deterministic optimal strategy which at any round t would pull an arm from $\mathcal{H}_{a^*}(t)$ if $|\mathcal{H}_{a^*}(t)| \geq 1$. $\square$

We call a deterministic optimal strategy that follows Lemmas C.6 and C.8 a *synchronizing strategy*. Such a strategy always alternates between two kinds of pulls: *gain pulls* where the strategy pulls from $\mathcal{P}_{a^*}(t) \setminus \mathcal{H}_{a^*}(t)$ and *synchronizing pulls* where the strategy pulls from $\mathcal{H}_{a^*}(t)$ and the maximum generative mean remains constant. We will now construct the *reduced* state transition graph. Before doing that for each arm $i \in [k]$ we introduce the following function p_i^{-1} defined as follows: (1) if $\tilde{\mu} \geq \mu_{i,\text{exit}}$ then $p_i^{-1}(\tilde{\mu}, n_i) = 0$, (2) if $\tilde{\mu}_i(p = n_i) > \tilde{\mu}$ then $p_i^{-1}(\tilde{\mu}, n_i) = 0$, and (3) otherwise: $p_i^{-1}(\tilde{\mu}, n_i) = m = \arg\max_{r \in \mathbb{Z}_{\geq 0}} \tilde{\mu}_i(p = n_i + r) \leq \tilde{\mu}$. Essentially given

some value for a generative mean $\tilde{\mu}$ and the current number of pulls arm i has received $p_i^{-1}(\tilde{\mu}, n_i)$ gives the maximum number of additional pulls m that arm i can receive such that the new value of i 's generative mean does not exceed $\tilde{\mu}$, i.e., $\tilde{\mu}_i(p = n_i + m) \leq \tilde{\mu}$. On the other hand, if $\tilde{\mu}_i(p = n_i) \geq \tilde{\mu}$ (i already has a higher generative mean) or $\tilde{\mu} \geq \mu_{i,\text{exit}}$ (the value of the generative mean exceeds i 's the exit mean) then $p_i^{-1}(\tilde{\mu}, n_i) = 0$.

Algorithm 2 FORMREDUCEDDAG

- 1: Draw the *first node* ($n_1^1 = p_1^{-1}(\tilde{\mu}_{a_1^*}, 0)$, $n_2^1 = p_2^{-1}(\tilde{\mu}_{a_2^*}, 0)$, $\dots$, $n_k^1 = p_k^{-1}(\tilde{\mu}_{a_k^*}, 0)$).
- 2: Draw a *gain node* (i, r) for each $i \in [k]$ and $r \in [T]$.
- 3: For each gain node (i, r) set $n_i^{(i,r)} = r$ and $\forall j \in [k], j \neq i$: $n_j^{(j,r)} = p_j^{-1}(\tilde{\mu}_i(p = r), 0)$.
- 4: Draw an edge from the first node to every node (i, r) only if $\mu_{i,\text{exit}} > \tilde{\mu}_{a_t^*}$ and $n_i^{(i,r)} = n_i^1 + 1$. Further, set the edge weight to $\sigma(\mu_i, \tilde{\mu}_{a_t^*}) - c_i + \sum_{\ell \neq i} [\sigma(\mu_\ell, \tilde{\mu}_{a_\ell^*}) - c_\ell] \cdot p_\ell^{-1}(\tilde{\mu}_i(p = r), n_\ell^1)$.
- 5: Draw an edge from every node (i, r) to (j, r') only if $\mu_{j,\text{exit}} > \tilde{\mu}_i(p = r)$ and $n_j^{(j,r')} = n_j^{(i,r)} + 1$. Set edge weight to $\sigma(\mu_j, \tilde{\mu}_i(p = r)) - c_j + \sum_{\ell \neq j} [\sigma(\mu_\ell, \tilde{\mu}_j(p = r')) - c_\ell] \cdot p_\ell^{-1}(\tilde{\mu}_j(p = r'), n_\ell^{(i,r)})$.
- 6: **return** Graph

The guarantees of the theorem follow immediately.